

Problem 1. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions of sets.

- (a) Prove that if $g \circ f$ is injective then f is injective.
- (b) Prove that if $g \circ f$ is surjective then g is surjective.
- (c) Find an example of $f : A \rightarrow B$ and $g : B \rightarrow A$ such that $f \circ g = \text{id}_B$ but neither f nor g is bijective.

Proof. **(a)** Suppose $g \circ f$ is injective but that f is not injective. So there exists $a, \alpha \in A$ such that $f(a) = f(\alpha)$ but $a \neq \alpha$. But then $g(f(a)) = g(f(\alpha))$ and $a \neq \alpha$, which contradicts that $g \circ f$ is injective. So f must be injective.

(b) Suppose $g \circ f$ is surjective but that g is not surjective. So there exists $c \in C$ such that for all $b \in B$, $g(b) \neq c$. But then for all $a \in A$, $g(f(a)) \neq c$, which contradicts that $g \circ f$ is surjective. So g must be surjective.

(c) Let $A = \{a\}$, $B = \{b, \beta\}$, $C = \{c\}$. Then let $f := a \mapsto b$ and $g := \{b, \beta\} \mapsto c$. Then f is not surjective, and g is not injective but $g \circ f$ is bijective via $(g \circ f)^{-1} := c \mapsto a$.

□

Problem 2. Let R be a commutative ring, and $S \subseteq R$ be a multiplicative set.

(a) Review the definition of $S^{-1}R$, and prove that $+$ and $\cdot$ make $S^{-1}R$ into a commutative ring with identity $1_R/1_R$.

(b) If $0 \in S$, show that $S^{-1}R$ is the zero ring.

Definition 1 (Localization). Let R be a commutative ring and $S \subseteq R$ a multiplicative submonoid; S contains 1_R and is closed under multiplication. The **localization** $S^{-1}R$ of R at S is defined as follows.

$$S^{-1}R := \left\{ \frac{r}{s} \mid r \in R, s \in S \right\} / \sim \text{ where } \frac{a}{b} \sim \frac{c}{d} \iff \exists s \in S \text{ such that } s(ad - bc) = 0.$$

We define $+$ and $\cdot$ on $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$ as follows:

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd}.$$

Also, notice that $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} = \frac{ca}{db} = \frac{c}{d} \cdot \frac{a}{b}$ and $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} = \frac{cb+da}{db} = \frac{c}{d} + \frac{a}{b}$. Commutativity is immediate.

We now briefly prove a lemma.

Lemma 1. If $\circ$ is a commutative binary operation on $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$ such that $x \circ y \sim x \circ y'$ for any x and any $y \sim y'$ in $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$, then for any $x \sim x', y \sim y'$ in $(x \circ y) \sim (x \circ y') \sim (x' \circ y) \sim (x' \circ y')$. That is: $[x]_{\sim} \circ [y]_{\sim} = [x \circ y]_{\sim}$

Proof. $x \circ y \sim x \circ y'$ for any $y \sim y' \implies y \circ x \sim y \circ x'$ if $x \sim x'$. Similarly, $x' \circ y' \sim x' \circ y$ if $y' \sim y \implies y' \circ x' \sim y' \circ x$ if $x' \sim x$. So then since $x \sim x'$ and $y \sim y'$, we have that $(x \circ y) \sim (x \circ y') \sim (x' \circ y) \sim (x' \circ y')$ by transitivity. $\square$

Next prove that $+, \cdot$ are well-defined on $S^{-1}R$.

Lemma 2. $+, \cdot$ are well-defined on $S^{-1}R$.

Proof. (a) We prove that $S^{-1}R$ is a commutative ring with identity 1. To begin we show that these $+, \cdot$ are well-defined on $S^{-1}R$. For any $x = \frac{a_1}{b_1} \sim \frac{a_2}{b_2} = x'$ and $y = \frac{c_1}{d_1} \sim \frac{c_2}{d_2} = y'$ in $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$, there exists $s, t \in S$ such that $s(a_1b_2 - b_1a_2) = 0$ and $t(c_1d_2 - d_1c_2) = 0$. Behold.

$$\begin{aligned} x + y &= \frac{a_1}{b_1} + \frac{c_1}{d_1} = \frac{a_1d_1 + b_1c_1}{b_1d_1}, \quad x + y' = \frac{a_1}{b_1} + \frac{c_2}{d_2} = \frac{a_1d_2 + b_1c_2}{b_1d_2} \\ x \cdot y &= \frac{a_1}{b_1} \cdot \frac{c_1}{d_1} = \frac{a_1c_1}{b_1d_1}, \quad x \cdot y' = \frac{a_1}{b_1} \cdot \frac{c_2}{d_2} = \frac{a_1c_2}{b_1d_2} \\ \implies (i) \quad &t(b_1d_2(a_1d_1 + b_1c_1) - b_1d_1(a_1d_2 + b_1c_2)) = t(\cancel{a_1b_1d_1d_2} + b_1^2c_1d_2 - \cancel{a_1b_1d_1d_2} - b_1^2d_1c_2) \\ &= tb_1^2(c_1d_2 - d_1c_2) = b_1^2(t(c_1d_2 - d_1c_2)) = b_1^2 \cdot 0 \implies x + y \sim x + y' \\ \implies (ii) \quad &t(b_1d_2(a_1c_1) - b_1d_1(a_1c_2)) = t(a_1b_1c_1d_2 - a_1b_1d_1c_2) = ta_1b_1(c_1d_2 - d_1c_2) \\ &= a_1b_1(t(c_1d_2 - d_1c_2)) = a_1b_1 \cdot 0 = 0 \implies x \cdot y \sim x \cdot y' \end{aligned}$$

By Lemma 1, since $+, \cdot$ are both commutative binary operations on $\left\{ \frac{r}{s} \mid r \in R, s \in S \right\}$ such that $x \circ y \sim x \circ y'$ for any x and any $y \sim y'$, we have that $[x]_{\sim} \circ [y]_{\sim} = [x \circ y]_{\sim}$ for $\circ = +, \cdot$. So $+$ and $\cdot$ are well-defined on $S^{-1}R$. $\square$

We finally address the problem on the next page.

Proof. (a) We now show that $S^{-1}R$ is a commutative ring with unity. For the sake of hygiene, we just use $\frac{a}{b}$ to denote any $[\frac{a}{b}]_{\sim}$ in $S^{-1}R$, since $+$, $\cdot$ are well-defined on it. Behold. For any $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \in S^{-1}R$, we show each of the ring axioms:

$$(\text{Associative } +) \frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f} \right) = \frac{a}{b} + \frac{cf + de}{df} = \frac{adf + bcf + deb}{bdf} = \left(\frac{a}{b} + \frac{c}{d} \right) + \frac{e}{f}$$

$$(\text{Associative } \cdot) \frac{a}{b} \cdot \left(\frac{c}{d} \cdot \frac{e}{f} \right) = \frac{a}{b} \cdot \frac{ce}{df} = \frac{ace}{bdf} = \left(\frac{a}{b} \cdot \frac{c}{d} \right) \cdot \frac{e}{f}$$

$$(\text{Commutative } +) \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} = \frac{cb + da}{db} = \frac{c}{d} + \frac{a}{b} \quad (*\text{Also shown in Definition 1*})$$

$$(\text{Commutative } \cdot) \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} = \frac{ca}{db} = \frac{c}{d} \cdot \frac{a}{b} \implies 1_{S^{-1}R} = \frac{1}{1} \quad (*\text{Also shown in Definition 1*})$$

$$\text{Also, for any } \frac{x}{s} \in S^{-1}R, s(x1 - 1x) = s(x - x) = 0, \forall s \in S, \implies \frac{x}{s} = 1_{S^{-1}R}, \forall x \in R$$

$$(1_{S^{-1}R}) \frac{1}{1} \cdot \frac{a}{b} = \frac{a}{b} \cdot \frac{1}{1} = \frac{a}{b}$$

$$(0_{S^{-1}R}) \frac{a}{b} + \frac{0}{1} = \frac{0}{1} + \frac{a}{b} = \frac{0(b) + 1a}{1b} = \frac{1a}{1b} = \frac{1}{1} \frac{a}{b} = 1_{S^{-1}R} \frac{a}{b} = \frac{a}{b}$$

$$\text{Also for any } s \in S, s^*(s(0) - (0)1) = s^*(0) = 0, \forall s^* \in S \implies \frac{0}{1} = \frac{0}{s} = 0_{S^{-1}R}, \forall s \in S$$

$$(\text{Additive Inverse}) \frac{a}{b} + \frac{-a}{b} = \frac{-a}{b} + \frac{a}{b} = \frac{-ab + ab}{bb} = \frac{0}{bb} = 0_{S^{-1}R}$$

$$(\text{Distributive}) \frac{a}{b} \cdot \left(\frac{c}{d} + \frac{e}{f} \right) = \frac{a}{b} \cdot \frac{cf + de}{df} = \frac{acf + ade}{bdf} = \frac{ac}{bd} + \frac{ae}{bf} = \left(\frac{a}{b} \cdot \frac{c}{d} \right) + \left(\frac{a}{b} \cdot \frac{e}{f} \right)$$

$$(\text{Distributive}) \left(\frac{a}{b} + \frac{c}{d} \right) \cdot \frac{e}{f} = \frac{ad + bc}{bd} \cdot \frac{e}{f} = \frac{ade + bce}{bdf} = \frac{ae}{bf} + \frac{ce}{df} = \left(\frac{a}{b} \cdot \frac{e}{f} \right) + \left(\frac{c}{d} \cdot \frac{e}{f} \right)$$

So $S^{-1}R$ is a commutative ring with $1_{S^{-1}R} = \frac{1}{1}$.

(b) If $0 \in S$, then for any $\frac{a}{b} \in S^{-1}R$, $\exists 0 \in S$ such that $0(a1 - 0b) = 0(a) = 0$ and so $\frac{a}{b} = \frac{0}{1} = 0_{S^{-1}R}, \forall \frac{a}{b} \in S^{-1}R$. So $S^{-1}R = \{0_{S^{-1}R}\}$ is the zero ring.

□

Problem 3. Let R be a commutative ring. Recall that we say R is Noetherian if every ideal is finitely generated. Prove that R is Noetherian if and only if every chain of ideals eventually stabilizes, i.e., if

$$I_0 \subseteq I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

is a chain of ideals, then $I_N = I_{N+1} = I_{N+k}$ for all $k > 0$ and some $N \in \mathbb{N}$.

Proof. ($\implies$) If R is Noetherian and $I_0 \subseteq I_1 \subseteq \cdots \subseteq I_n \subseteq \cdots$ is a countable chain of ideals in R , consider $I = \bigcup_{i \in \mathbb{N}} I_i$. Behold. $\forall a, b \in I$

$$(\leq_+) : a, b \in I \implies \exists i, j \in \mathbb{N} \text{ such that } a \in I_i, b \in I_j \implies \text{Without loss of generality } i \leq j$$

$$\implies a, b \in I_j \leq_+ R \implies a - b \in I_j \subseteq I \implies I \leq_+ R$$

$$(\text{Absorption}) : a \in I \implies \exists i \in \mathbb{N} \text{ such that } a \in I_i$$

$$\implies \forall r \in R, ra = ar \in I_i \subseteq I \implies \forall r \in R, \forall a \in I, ra = ar \in I.$$

So I is an ideal of R , and since R is Noetherian, I is finitely generated. So there exist $a_1, \dots, a_n \in I$ such that $I = \langle a_1, \dots, a_n \rangle$. Now, for each $i \in [n]$, $a_i \in I \implies \exists k_i \in \mathbb{N}$ such that $a_i \in I_{k_i} \subseteq I$. Let $N = \max\{k_1, \dots, k_n\}$. Then $I_{k_i} \subseteq I_N, \forall i \in [n] \implies a_1, \dots, a_n \in I_N \subseteq I = \langle a_1, \dots, a_n \rangle \implies I_N = I$. So then

$$I_1 \subseteq \cdots \subseteq I = I_N \subseteq I_{N+1} \subseteq \cdots \subseteq I \implies I_k = I, \forall k \geq N.$$

($\impliedby$) On the other hand if every countable ideal chain stabilizes, consider any ideal I of R . If $I = \langle 0 \rangle$ or $I = \langle 1 \rangle$ it is finitely generated. Otherwise, pick some $a_1 \in I$. If $i = \langle a_i \rangle$, it is finitely generated. Otherwise pick some $a_2 \in I \setminus \{a_1\}$, if $I = \langle a_1, a_2 \rangle$ it is finitely generated. If we continue in this fashion, we build a countable chain of ideals each contained in I :

$$\langle 0 \rangle \subseteq \langle a_1 \rangle \subseteq \langle a_1, a_2 \rangle \subseteq \cdots \subseteq I$$

Since every countable ideal chain stabilizes, there exists some $N \in \mathbb{N}$ such that $\langle a_1, \dots, a_N \rangle = \langle a_i \mid 1 \leq i \leq k \rangle, \forall k \geq N$. So I is finitely generated. That is, this process must terminate after picking some $N \in \mathbb{N}$ such that $\langle a_1, \dots, a_N \rangle = I$.

□

Problem 4. Let R be a commutative ring, let I_j be ideals of R , and let P be a prime ideal of R .

- (a) Prove that if $I_1 \cdots I_n \subseteq P$, then $I_j \subseteq P$ for some j . **Hint:** Use contradiction.
- (b) By (a), if $P \supseteq \bigcap_{j=1}^r I_j$, then P contains one of the ideals I_j . Prove or disprove: if $P \supseteq \bigcap_{j=1}^{\infty} I_j$, then P contains one of the I_j .

Proof. (a) $I_1 \subseteq P \implies I_1 \subseteq P$. Now assume that for some $k \geq 1$, $I_1 \cdots I_k \subseteq P \implies I_j \subseteq P$ for some $j \in [k]$. Next, if $I_1 \cdots I_{k+1} \subseteq P$, suppose that $I_{k+1} \not\subseteq P$ and $I_1 \cdots I_k \not\subseteq P$. Then $\exists (a_1 \cdots a_k) \in I_1 \cdots I_k$ and $\exists a_{k+1} \in I_{k+1}$ such that $(a_1 \cdots a_k), a_{k+1} \notin P$. But then $(a_1 \cdots a_k)a_{k+1} \in I_1 \cdots I_{k+1} \subseteq P \implies P$ is not prime, a contradiction. So either $I_{k+1} \subseteq P$ or $I_1 \cdots I_k \subseteq P \implies I_j \subseteq P$ for some $j \in [k]$. So $I_1 \cdots I_{k+1} \subseteq P \implies I_j \subseteq P$ for some $j \in [k+1]$. Finally, by induction, $I_1 \cdots I_n \subseteq P \implies I_j \subseteq P$ for some $j \in [n]$.

(b) We prove the finite intersection result. For any $(a_1 \cdots a_n) \in I_1 \cdots I_n$ and then any $k \in [n]$, notice that $a_1 \cdots a_n = (a_1 \cdots a_{k-1})a_k(a_{k+1} \cdots a_n) = (a_1 \cdots a_{k-1})(a_{k+1} \cdots a_n)a_k \in I_k$. So then $(a_1 \cdots a_n) \in I_k$, $\forall k \in [n] \implies (a_1 \cdots a_n) \in \bigcap_{i=1}^n I_i \implies I_1 \cdots I_n \subseteq \bigcap_{i=1}^n I_i \subseteq P$. Therefore, by 4(a), $I_1 \cdots I_n \subseteq P \implies I_1 \cdots I_n \subseteq P \implies I_j \subseteq P$ for some $j \in [n]$.

Now consider the countably infinite collection of ideals $\{\langle x^i \rangle \in \mathcal{P}(k[[x]]) \mid i \in \mathbb{Z}^+\}$ of $R = k[[x]]$ for some unital commutative ring k . $I = \bigcap_{i=1}^{\infty} \langle x^i \rangle$. Any element of I is of the form rx^j for some $r \in k$ and some $j \geq 1$. Additionally, $rx^j \in I \subseteq \langle x^{j+1} \rangle \implies rx^j = r'x^{j+1}$ for some $r' \in k$. Therefore by the definition of equality in $k[[x]]$, $r = r' = 0$ and in fact $I = \bigcap_{i=1}^{\infty} \langle x^i \rangle = \{0\}$, a prime ideal. However, $\langle x^i \rangle \not\subseteq \{0\}, \forall i \geq 1$, and we have a counterexample in this intersection of ideals in $R = k[[x]]$.

□

Problem 5. Let R be a commutative ring, and let $\text{nil}(R)$ be the set of nilpotents of R , i.e.

$$\text{nil}(R) := \{r \in R \mid r^n = 0 \text{ for some } n \in \mathbb{N}\}.$$

This is often called the nilradical of R , and you proved in the last homework that it is an ideal of R . Prove that $\text{nil}(R)$ is contained in every prime ideal of R .

Proof. Let P be any prime ideal of R , and consider any $x \in \text{nil}(R)$. If $x^0 = 1 = 0$, then $R = \{0\}$ and so $x = 0 \in P$. If $x^1 = x = 0 \in P \leq_+ R$, then $x \in P$. Suppose $x^k \in P \implies x \in P$ for some $k \geq 1$. Then if $x^{k+1} = x(x^k) \in P$, since P is prime, $x \in P$ or $x^k \in P \implies x \in P$. So then by induction $x^n \in P \implies x \in P$ for all $n \geq 1$. Therefore, $\forall x \in \text{nil}(R), \exists n \in \mathbb{N}$ such that $x^n = 0 \in P \implies x \in P$. So $\text{nil}(R) \subseteq P$. Thus, $\text{nil}(R)$ is contained in any prime ideal P of R .

□

Problem 6. A ring is called local if it has a unique maximal ideal. Let R be a commutative local ring. Prove that if $x \in R$, then either x or $1 - x$ is a unit.

Proof. Let M be the unique maximal ideal of R . Since M is proper, it can't contain 1 otherwise $1r = r1 = r \in M, \forall r \in R$. Next, if $y \in R$ is not a unit, then $ry = yr \neq 1, \forall r \in R$ and so $\langle y \rangle = \{ry \in R \mid r \in R\}$ doesn't contain 1. That is, $\langle y \rangle \subset R = \langle 1 \rangle$. So then $\langle y \rangle = M$ or $\langle y \rangle \subset M$ since M is unique and maximal. In either case $y \in M$. So M contains all non-units of R .

Now, suppose x and $1 - x$ are both non-units in R . Then $x, 1 - x \in M$. But then $x + 1 - x = 1 \in M$, a contradiction. So either x or $1 - x$ is a unit in R . $\square$

Problem 7. Let R be a commutative ring, and let $I \subseteq R$ be a proper ideal. Prove that the set of prime ideals containing I has minimal elements. These are the minimal primes of I . **Hint:** Use Zorn's Lemma.

Proof. Let S be the set of prime ideals of R that contain I . In **Homework 2** we proved that the set of all proper ideals containing I is a poset with set inclusion. So then $(S \setminus \{R\}, \subseteq)$ is an induced subposet of that poset. Obviously, $P \subseteq R, \forall P \in S \setminus \{R\}$ and so $(S, \subseteq)$ is also a poset. Now consider any chain C in S .

Let $L_C = \bigcap_{P \in C} P$. Obviously, $L \subseteq C, \forall P \in C \implies I \subseteq L$. Next, $\forall a, b \in L, a, b \in P, \forall P \in C \implies a - b \in P, \forall P \in C$ since they are all additive subgroups of R . So $L \leq_+ R$. Also, $\forall r \in R, \forall a \in L, ra \in P, \forall P \in C \implies ra = ar \in L$. So L is an ideal of R containing I . Finally, if $ab \in L$, then $ab \in P, \forall P \in C \implies a$ or $b \in P, \forall P \in C \implies a$ or $b \in L$. So L is a prime ideal of R containing I , and a lower bound for C in S . So every chain in S has a lowerbound.

So then in the dual poset S^* every chain has an upperbound in S , and by Zorn's Lemma, there exists a maximal element in S^* . That is, there exists a minimal element in S . $\square$

Problem 8. Let $\mathbb{K}$ be a field. Prove that $\mathbb{K}$ contains a unique smallest subfield $\mathbb{k}$ and that $\mathbb{k}$ is isomorphic to either $\mathbb{Q}$ or $\mathbb{Z}/p\mathbb{Z}$ for some prime p . $\mathbb{k}$ is called the prime subfield of F . **Hint:** Intersect all subfields of $\mathbb{K}$.

Proof. Let $\mathcal{S}(K)$ be the set of all subfields of $\mathbb{K}$ and $\mathbb{k} = \bigcap_{\mathbb{F} \in \mathcal{S}(K)} \mathbb{F}$. Obviously, $\mathbb{F} \subseteq \mathbb{K}$, $\forall \mathbb{F} \in \mathcal{S}(K) \implies \mathbb{k} \subseteq \mathbb{K}$. We show $\mathbb{k}$ is a subfield of $\mathbb{K}$. $\forall x, y \in \mathbb{k}$, $x, y \in \mathbb{F}$, $\forall \mathbb{F} \in \mathcal{S}(K)$ and

$$x + y \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies x + y \in \mathbb{k}$$

$$xy \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies xy \in \mathbb{k}$$

$$-x \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies -x \in \mathbb{k}$$

$$x^{-1} \in \mathbb{F}, \forall \mathbb{F} \in \mathcal{S}(K) \implies x^{-1} \in \mathbb{k}$$

Then, $\mathbb{k} = \bigcap_{\mathbb{F} \in \mathcal{S}(K)} \mathbb{F} \subseteq \mathbb{F}$, $\forall \mathbb{F} \in \mathcal{S}(K) \implies \mathbb{k}$ is the smallest subfield of $\mathbb{K}$. Well, for any other minimal element $\mathbb{k}' \in \mathcal{S}(\mathbb{K})$, $\mathbb{k}' \subseteq \mathbb{k}$ and $\mathbb{k} \subseteq \mathbb{k}' \implies \mathbb{k} = \mathbb{k}'$. So $\mathbb{k}$ is the unique smallest subfield of $\mathbb{K}$. Next, let

$$n_+ = \sum_{i=1}^n 1_{\mathbb{k}}, \forall n \in \mathbb{Z}^+ \text{ and } 0_+ = 0.$$

If $\text{Char } \mathbb{k} = 0$, then let $\phi : \mathbb{Q} \rightarrow \mathbb{k}$ be defined via $\phi(\frac{a}{b}) = \frac{a_+}{b_+} = a_+ b_+^{-1}$. This is well-defined since (i) $b \neq 0 \implies b_+ \neq 0_{\mathbb{k}} \implies b_+^{-1} \in \mathbb{k}$. Also, (ii) $\frac{a}{b} = \frac{c}{d} \implies ad = bc \implies a_+ d_+ = b_+ c_+ \implies \frac{a_+}{b_+} = \frac{c_+}{d_+}$. We show this is a ring homomorphism. For any $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$,

$$\phi(1) = 1_+$$

$$\phi\left(\frac{a}{b} + \frac{c}{d}\right) = \phi\left(\frac{ad + bc}{bd}\right) = \frac{(ad + bc)_+}{(bd)_+} = \frac{a_+ d_+ + b_+ c_+}{b_+ d_+} = \frac{a_+ d_+}{b_+ d_+} + \frac{b_+ c_+}{b_+ d_+}$$

$$= \frac{a_+}{b_+} \cdot \frac{d_+}{d_+} + \frac{c_+}{d_+} \cdot \frac{b_+}{b_+} = \phi\left(\frac{a}{b}\right) + \phi\left(\frac{c}{d}\right)$$

$$\phi\left(\frac{a}{b} \cdot \frac{c}{d}\right) = \phi\left(\frac{ac}{bd}\right) = \frac{(ac)_+}{(bd)_+} = \frac{a_+ c_+}{b_+ d_+} = \left(\frac{a_+}{b_+}\right) \left(\frac{c_+}{d_+}\right) = \phi\left(\frac{a}{b}\right) \cdot \phi\left(\frac{c}{d}\right)$$

Well, $\frac{a_+}{b_+} = \frac{c_+}{d_+} \implies a_+ d_+ = b_+ c_+ \implies ad = bc \implies \frac{a}{b} = \frac{c}{d}$, so ϕ is injective and $\ker \phi = \{0\}$. So then by the **First Isomorphism Theorem**, $\mathbb{Q} \cong \mathbb{Q}/\{0\} \cong \phi(\mathbb{k}) \subseteq \mathbb{k}$. Well, since $\phi(\mathbb{k}) \subseteq \mathbb{k} \subseteq \mathbb{K}$ must then be a field, $\phi(\mathbb{k}) = \mathbb{k}$ since otherwise $\mathbb{k}$ isn't minimal. So $\mathbb{Q} \cong \mathbb{k}$.

If $\text{Char } \mathbb{k} = p > 0$, then consider $\mathbb{k}_p = \langle 1_+ \rangle_+ \leq_+ \mathbb{k}$. Obviously, since $\text{Char } \mathbb{k} = p$, $\mathbb{k}_p = \{0, 1_+, 2_+, \dots, (p-1)_+\}$. All non-zero elements of $\mathbb{k}_p$ are units, and so $\mathbb{k}_p \subseteq \mathbb{k} \subseteq \mathbb{K}$ is a field $\implies \mathbb{k}_p = \mathbb{k}$ since $\mathbb{k}$ is minimal. Well, then obviously $\phi := n_+ \mapsto n$ is a bijection from $\mathbb{k}$ to $\mathbb{Z}_p$. We show this is a ring homomorphism. For any $a_+, b_+ \in \mathbb{k}$,

$$\phi(1_+) = 1$$

$$\phi(a_+ + b_+) = a + b = \phi(a_+) + \phi(b_+)$$

$$\phi(a_+ b_+) = ab = \phi(a_+) \phi(b_+)$$

So then ϕ is a field isomorphism and $\mathbb{k} \cong \mathbb{Z}_p$.

□

Problem 9. Let P be a prime ideal of a commutative ring R , and let R_P be the localization of R at P .

- (a) Prove that if $f \notin P$, then $f/1$ is a unit in R_P .
 (b) Let $I \subseteq P$ be an ideal of R . Prove that

$$IR_P = \left\{ \frac{a}{b} \in R_P \mid a \in I \right\}$$

is a proper ideal of R_P .

- (c) Prove that if I is an ideal of R and $I \not\subseteq P$, then $IR_P = R_P$.
 (d) Prove that R_P is a local ring, and describe its maximal ideal.

Proof. **(a)** For a prime ideal P of R , the localization of R at P is $R_P = (R \setminus P)^{-1}R$ and just note that these fractions are equivalence classes. Now, if $f \notin P$ then $\frac{f}{1} \in R_P$ and so $\frac{1}{f} \cdot \frac{f}{1} = \frac{f}{1} \cdot \frac{1}{f} = \frac{1}{1} = 1_{R_P}$. So $\frac{f}{1}$ is a unit in R_P .

(b) If I is an ideal of R , Consider any $\frac{a}{b}, \frac{c}{d} \in IR_P$. $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$. Suppose $bd \in P$. But then since P is prime, $b \in P$ or $d \in P$, a contradiction. So $bd \notin P$. Also, $ad-bc \in I$ since I is an ideal of R . So then $\frac{ad-bc}{bd} \in IR_P \leq_+ R_P$. Now, for any $\frac{r}{q} \in R_P$ and any $\frac{i}{b} \in IR_P$, $\frac{i}{b} \cdot \frac{r}{q} = \frac{r}{q} \cdot \frac{i}{b} = \frac{ri}{qb}$. Once more, $qb \in P \implies q \in P$ or $b \in P$, a contradiction. So $qb \notin P$ and since $ri \in I$, $\frac{ri}{qb} \in IR_P$ which must then be an ideal of R_P .

If $I \subseteq P$, IR_P isn't proper. Then $1_{R_P} = \frac{1}{1} \in IR_P \implies 1 \in I \subseteq P$. But then $1 \notin (R \setminus P)$, and so $\frac{1}{1} \notin R_P$ in the first place, a contradiction. So $I \not\subseteq P \implies IR_P$ is a proper ideal of R_P .

(c) We simply prove the other direction ($\supseteq$) continuing after proving $IR_P \subseteq R_P$ is an ideal in the first part of **(b)**, with the added assumption that $I \not\subseteq P$. Since any $i \in I$ does not belong to P , for any $\frac{i}{q} \in IR_P$, $\frac{q}{i} \in R_P \implies \frac{q}{i} \cdot \frac{i}{q} = \frac{i}{q} \cdot \frac{q}{i} = \frac{1}{1} = 1_{R_P} \in IR_P$ since it is an ideal of R_P and so $IR_P = R_P$.

(d) Consider any proper ideal $I_P \subset R_P$. Since it can't contain 1_{R_P} , none of its elements can be units. That is, they can't be of the form $\frac{f}{1}$ where $f \notin P$ by **(a)**. So then

$$I_P \subseteq \left\{ \frac{a}{b} \in R_P \mid a \in P \right\} \implies I_P = \left\{ \frac{a}{b} \in R_P \mid a \in I \subseteq P \right\}.$$

Well, since I_P is an ideal, for any such $\frac{a}{b} \in I_P$, and any $\frac{r}{q} \in R_P$, we must have that $\frac{a}{b} \cdot \frac{r}{q} = \frac{r}{q} \cdot \frac{a}{b} = \frac{ra}{qb} \in I_P$ and that $\frac{a}{b} - \frac{r}{q} = \frac{aq-br}{bq} \in I_P$. Which means that $ra = ar \in I, \forall r \in R, \forall a \in I$ and that $\frac{a}{b} - \frac{r}{q} \in I_P, \forall r \in R, \forall q \in R \setminus P, \forall r \in R, \forall b, q \in R \setminus P, \forall a \in I \implies \frac{a}{1} - \frac{b}{1} = \frac{a-b}{1} \in I_P, \forall a, b \in I \implies a-b \in I, \forall a, b \in I \implies I \leq_+ R$. That is, $I \subseteq P$ is an ideal of P . So any proper ideal $I_P \subset R_P$ is of the form $I_P = IP_P$ for some ideal $I \subseteq P$.

Well,

$$IR_P = \left\{ \frac{a}{b} \in R_P \mid a \in I \subseteq P \right\} \subseteq \left\{ \frac{a}{b} \in R_P \mid a \in \bigcup_{I \subseteq P} I \right\} = PR_P \text{ for any ideal } I \subseteq P \text{ of } R.$$

So PR_P contains all proper ideals of R_P . Thus, R_P is a local ring with the unique maximal ideal PR_P . □